



SECURITY CONTROL DETECTIVE

Find the Risk. Choose the Control. Save the Organization!

Course	ITEP 413 – Information Assurance and Security 2
Mode	Self-Paced / Individual
Suggested Time	60–90 minutes
Submission	Digital worksheet or LMS submission

Learning Objectives

1. Identify common business drivers for information security.
2. Distinguish physical, administrative/procedural, and technical controls.
3. Classify controls as preventive, detective, or corrective.
4. Select appropriate controls for a given security problem.
5. Explain how security policies and controls work together.
6. Identify ways to mitigate security risks.

Key idea: Policy explains the “why” and “what,” while controls explain the “how” and measurement.



MISSION 1: Identify the Security Problem

You have been hired as a Junior Information Security Analyst for TechNova Solutions. Identify the business driver most affected in each case.

Choose from: Cost • Customer Satisfaction • Compliance • Measurement

Case A: The company receives a large fine because employees failed to follow a government data-protection requirement.

Answer: Compliance

Case B: Customers stop using the company's online service because their personal information was exposed.

Answer: Customer Satisfaction

Case C: The company spends millions of pesos recovering from repeated security incidents because it has no effective backup or recovery process.

Answer: Cost

Case D: Management cannot determine how many employees have completed security-awareness training.

Answer: Measurement



Mission Score: 1 point each = 4 points



MISSION 2: Control Detective

Classify each control as Physical, Administrative/Procedural, or Technical.

Situation	Physical	Administrative	Technical
1. CCTV camera monitors the server room	■	☐	☐
2. Employees attend security-awareness training	☐	■	☐
3. Firewall blocks unauthorized network traffic	☐	☐	■
4. Security guard checks visitors	■	☐	☐
5. Manager approves access to confidential files	☐	☐	☐
6. Antivirus detects malicious software	☐	☐	■



Mission Score: 1 point each = 6 points



MISSION 3: Stop the Attack!

Classify each situation: Preventive = blocks threats; Detective = identifies breaches; Corrective = restores operations.

1. A firewall blocks unauthorized network connections.

Answer: Preventive

2. An intrusion detection system sends an alert when suspicious activity occurs.

Answer: Detective

3. A backup is used to restore files after a hard-drive failure.

Answer: Corrective

4. A user is required to enter an MFA code before accessing the system.

Answer: Preventive

5. Security personnel review logs to identify suspicious activity.

Answer: Detective

6. A company restores its database after ransomware damages the original data.

Answer: Corrective



Mission Score: 1 point each = 6 points



MISSION 4: Build the Perfect Defense

Scenario: A university stores student grades, personal information, and enrollment records in an online system.

- Several employees used weak passwords.

- Someone attempted to access the database without authorization.
- The university discovered that security logs were not being reviewed.
- A server failed and some files became unavailable.
- Employees were unsure how to handle confidential student information.

Choose at least ONE control for each problem.

Problem	Your Proposed Control	Control Type	Design Type
Weak passwords	Strong Password Policy + MFA	Administrative / Technical	Preventive
Unauthorized access attempt	Access controls/ auth + intrusion detection	Technical	Preventive / Detective
Logs not reviewed	Regular security log reviews	Administrative	Detective
Server failure	Regular backups	Technical	Corrective
Employees mishandle confidential data	Security-awareness training on handling confidential info	Administrative	Preventive

Example: Weak passwords → Strong password policy + MFA → Technical / Administrative → Preventive

MISSION 5: Policy or Control?

Determine whether each statement is primarily a POLICY or a CONTROL.

1. “Employees must protect confidential customer information.”

Answer: Policy

2. “The organization will use encryption when transmitting sensitive information.”

Answer: Control

3. “Only authorized personnel may access student records.”

Answer: Control

4. “Multi-factor authentication will be required for access to the student database.”

Answer: Control

 Mission Score: 1 point each = 4 points

MISSION 6: Privacy Challenge

A university registration form asks students for:

- Full name
- Student number
- Email
- Home address
- Favorite movie
- Favorite food
- Bank account password

A. Which information should NOT be collected?

Favorite movie, favorite food, bank account password

B. Why?

Universities should collect only informations that are necessary for providing the registration service.

C. Give TWO security controls that could protect students' PII.

Encryption and Access controls



Mission Score: 6 points



FINAL BOSS: Save TechNova!

TechNova has the following problems:

- No employee security training
- No firewall
- No backup system
- Confidential documents are not classified
- No process for reviewing security incidents

Create a 5-control security plan.

#	Security Problem	Recommended Control	Type	Preventive / Detective / Corrective
1	No employee security training	Security-awareness training program	Administrative	Preventive
2	No firewall	Install and configure a firewall	Technical	Preventive
3	No backup system	Regular data backups	Technical	Corrective
4	Confidential documents are not classified	Data classification policy and confidential labels	Administrative	Preventive
5	No process for reviewing security incidents	Regular security logs	Administrative	Detective

★ **Bonus Challenge:** Choose ONE of your controls and explain why it is important to the organization. Write 3–5 sentences.

Security-awareness training. It is important because employees need to understand their responsibilities when handling security risks. The material notes that employee negligence is a leading cause of breaches. Training helps employees recognize risks and follow security policies consistently. It therefore helps reduce the possibility of security incidents.

Scoring

Mission	Points
Mission 1 – Business Drivers	4
Mission 2 – Control Detective	6
Mission 3 – Stop the Attack	6
Mission 4 – Perfect Defense	10
Mission 5 – Policy or Control	4
Mission 6 – Privacy Challenge	6
Final Boss	10
Total	46
Bonus	+5

Performance Rating

-  41–46 = Security Master
-  35–40 = Security Analyst
-  28–34 = Junior Security Analyst
-  20–27 = Security Trainee
-  Below 20 = Retry the Mission

Student Submission

Submit one completed answer sheet containing:

- Mission 1 answers
- Mission 2 classifications
- Mission 3 classifications
- Mission 4 security-control table
- Mission 5 answers
- Mission 6 privacy answers
- Final Boss security plan
- Bonus explanation, if completed

Self-Paced Rule

Students may work independently and at their own pace. They may review the Week 3 presentation while completing the missions. The goal is not simply to memorize the terms, but to apply security controls to realistic situations.